



SQL ROADMAP

SQL

Step by Step to Become SQL Pro



Master SQL step by step. Practice consistently.
Work with real data. Build real skills.



1



SQL Basics

- What is SQL?
- Databases, Tables, Rows, Columns
- Data Types
- SELECT Statement

1–2
Days

OUTCOME

Understand database basics and retrieve data.



2



Filtering Data

- WHERE Clause
- Comparison Operators
- Logical Operators (AND, OR, NOT)
- IN, BETWEEN, LIKE, IS NULL

2–3
Days

OUTCOME

Filter data exactly the way you need.



3



Sorting & Limiting Data

- ORDER BY
- ASC / DESC
- LIMIT / TOP / FETCH

1–2
Days

OUTCOME

Sort and limit results effectively.



4



Aggregate Functions

- COUNT(), SUM(), AVG(), MIN(), MAX()
- Aggregate with WHERE
- GROUP BY Clause
- HAVING Clause

2–3
Days

OUTCOME

Summarize and analyze data like a pro.



5



Joins

- INNER JOIN
- LEFT JOIN / RIGHT JOIN
- FULL JOIN
- Self Join

3–4
Days

OUTCOME

Combine data from multiple tables.



6



Subqueries

- What is a Subquery?
- Subquery in SELECT
- Subquery in WHERE
- Correlated Subqueries

2–3
Days

OUTCOME

Write powerful queries using subqueries.



7



Advanced SQL

- CASE Statement
- COALESCE, NULLIF
- Window Functions (ROW_NUMBER, RANK, DENSE_RANK, LEAD, LAG)
- Common Table Expressions (CTE)

4–5
Days

OUTCOME

Solve complex problems with advanced SQL.



8



Database Design Basics

- Primary Key, Foreign Key
- Relationships (One-to-One, One-to-Many, Many-to-Many)
- Constraints (NOT NULL, UNIQUE, CHECK, DEFAULT)
- Normalization Basics

3–4
Days

OUTCOME

Design clean and efficient databases.



9



Performance Optimization

- Indexes
- Query Optimization Tips
- Execution Plan Basics

2–3
Days

OUTCOME

Write fast and optimized queries.



10



Practice & Real Projects

- Practice on real datasets
- Solve problems on different topics
- Build projects (e.g., Sales Analysis, HR System, E-commerce Reports)

Ongoing

OUTCOME

Build portfolio and become job ready.



POPULAR TOOLS



MySQL



PostgreSQL



SQL Server



SQLite



DBeaver

PRACTICE PLATFORMS



LeetCode – Database



HackerRank – SQL



Mode Analytics



SQLBolt



TIPS FOR SUCCESS



Practice Daily



Write More Queries



Work on Real Data



Understand, Don't Memorize



Practice is the key to mastery. Keep querying, keep growing!

